Weekly Student Progress Performa

Team Id	CS26084	**Week no / Date**	Week 11 27 September 2026
Project Title	SentinelCore: A Network Detection and Incident Response Platform		
Members Present	Om Rashiya (24cs084), Yash Patel (24cs082)		
Mentor	Akshita Kadam		

WEEK 11 • Reporting, Containment, PCAP, and Intelligence

Section	Details to be filled by the team
Planned Tasks (from last week)	Complete reporting, containment, PCAP analysis, and threat-intelligence enrichment.
Tasks Completed This Week	Implemented four asynchronous report types, report schedules, PDF/CSV/JSON outputs, protected and TTL-bounded firewall blocks, PCAP upload/parser isolation, flow/artifact extraction, IOC/source management, feed normalization, and event/incident enrichment.
Commits Made	Total: 5 milestone commits. 87f7e20 — M9 reporting backend; e536b79 — M9 reporting frontend; fe1ca48 — M10 firewall containment; 7082a69 — M5 to M10 work; 0c98a3e — all-module integration.
Issues Created / Closed	Created: Uploaded captures and external feeds are untrusted; firewall operations are high impact. Closed: Isolated parsing, bounded file/resource use, normalized IOCs, protected critical addresses, and enforced automatic unblock expiry.
Milestone Status	~91% complete — on track. All planned functional modules were integrated end to end.
Blockers / Challenges Faced	Safely parsing large/malformed captures and feeds while keeping memory, database work, and external requests bounded.
Plan for Next Week	Run end-to-end validation, harden security controls, improve documentation/UI clarity, and prepare the final demonstration.

Research and Technical Study

Research objective. Complete reporting, containment, PCAP analysis, and threat-intelligence enrichment.

Study completed. Reviewed PCAP parser isolation, MIME/size validation, flow reconstruction, IOC normalization, safe feed retrieval, request controls, and reversible containment. Indicators remain attributable to their source and match evidence.

Integrated module safeguards

Module	Untrusted or high-impact input	Implemented safeguard
Reporting	Long-running generation	Queued jobs and bounded status flow
Firewall	Target and duration	Protected addresses plus mandatory TTL
PCAP	Uploaded capture	Magic bytes, size bounds, isolated parsing
Threat intelligence	IOC and source data	Normalization, attribution, request controls
Investigation	Cross-module evidence	Links to incident, event, flow, and IOC records

Decision carried into the project

Implemented four asynchronous report types, report schedules, PDF/CSV/JSON outputs, protected and TTL-bounded firewall blocks, PCAP upload/parser isolation, flow/artifact extraction, IOC/source management, feed normalization, and event/incident enrichment.

Implementation Record and Website Evidence

Reporting

Implemented incident summary, incident detail, asset inventory, and event statistics generation with queued status tracking, supported output formats, downloads, and scheduled execution.

Guarded containment

Implemented admin-only blocks with required reason and TTL, protected gateway/DNS/platform targets, helper-mediated apply/revoke operations, expiry processing, reconciliation, and retained history.

PCAP analysis

Added bounded upload validation, capture metadata, isolated parser execution, flow and artifact extraction, packet/stream views, filtering, and the ability to attach evidence to incidents.

Threat intelligence

Added normalized IOCs, administrator-managed sources, matching against events and PCAP artifacts, retro-hunts, match history, and enrichment links into investigation views.

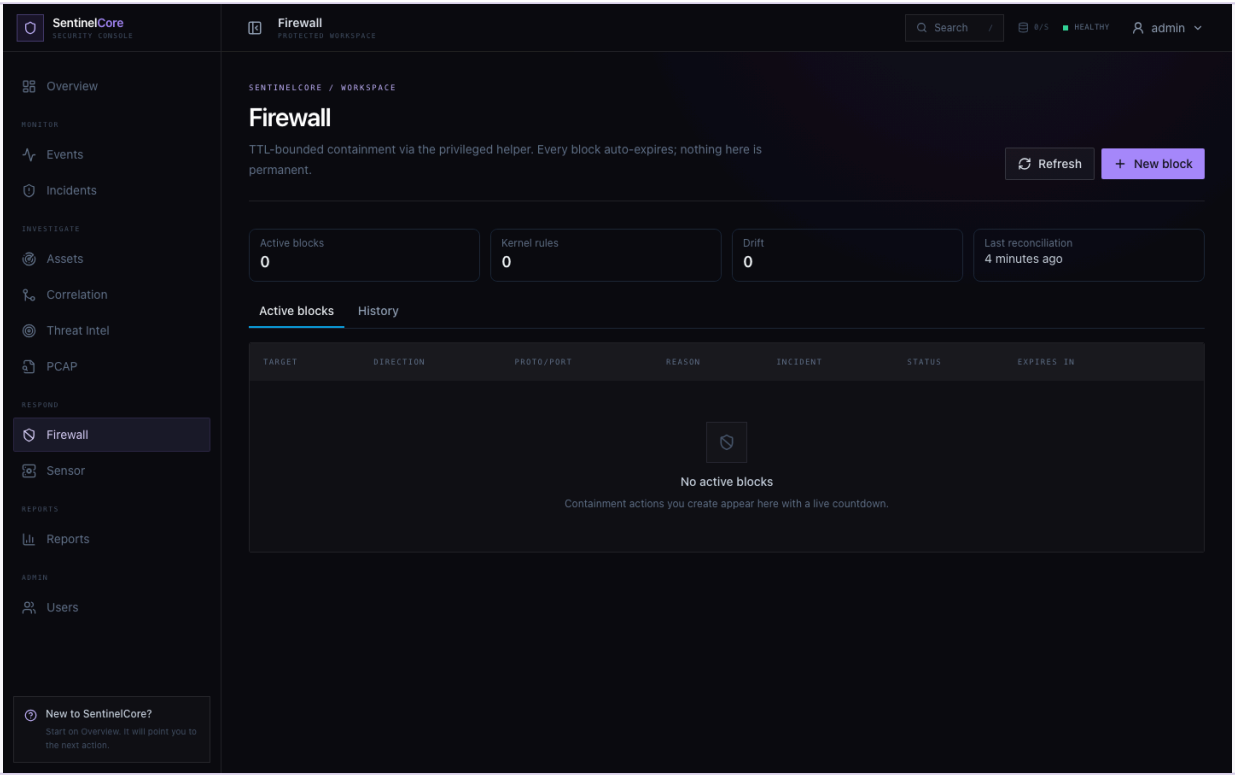


Figure W11.1 — Project-owned SentinelCore interface mapped specifically to Week 11.

Evidence interpretation. The Firewall workspace shows active-block, kernel-rule, drift, and reconciliation-age summaries, plus New block, Active blocks, History, and expiry-oriented containment controls. Its visible controls, labels, status fields, and navigation provide direct evidence for the work described on this page.

Repository and Architecture Evidence

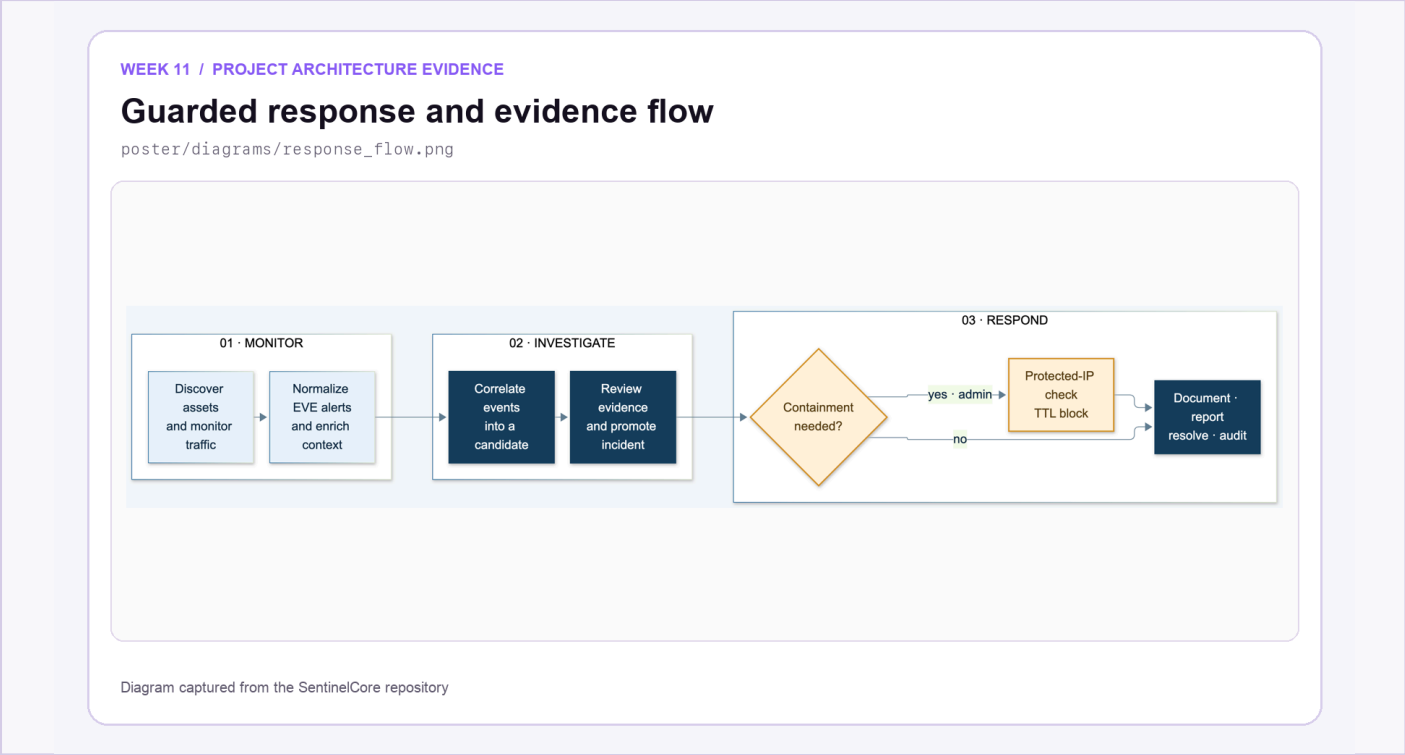


Figure W11.2 — Guarded response and evidence flow. This is a separate internal artifact, not a copy of the website image.

Repository files used for traceability

- reports/modules/M09-reporting_1.md
- reports/modules/M10-firewall_1.md
- reports/modules/M11-pcap_1.md
- reports/modules/M12-threat-intel_1.md
- frontend/src/pages/firewall/FirewallPage.jsx

Verification

All named report, firewall, PCAP, and intelligence capabilities were checked against their module specifications, pages, routes, tests, and milestone commits.

Traceability conclusion. Guarded response and evidence flow directly supports the Reporting, Containment, PCAP, and Intelligence milestone. All named report, firewall, PCAP, and intelligence capabilities were checked against their module specifications, pages, routes, tests, and milestone commits. The artifact, source paths, implementation record, and website evidence therefore describe the same weekly scope.

Weekly Closure and Faculty Assessment

Completed outcome

Implemented four asynchronous report types, report schedules, PDF/CSV/JSON outputs, protected and TTL-bounded firewall blocks, PCAP upload/parser isolation, flow/artifact extraction, IOC/source management, feed normalization, and event/incident enrichment.

Commit record

Total: 5 milestone commits. 87f7e20 — M9 reporting backend; e536b79 — M9 reporting frontend; fe1ca48 — M10 firewall containment; 7082a69 — M5 to M10 work; 0c98a3e — all-module integration.

Issue and resolution

Created: Uploaded captures and external feeds are untrusted; firewall operations are high impact. Closed: Isolated parsing, bounded file/resource use, normalized IOCs, protected critical addresses, and enforced automatic unblock expiry.

Milestone position

~91% complete — on track. All planned functional modules were integrated end to end.

Challenge handled

Safely parsing large/malformed captures and feeds while keeping memory, database work, and external requests bounded.

Next-week hand-off

Run end-to-end validation, harden security controls, improve documentation/UI clarity, and prepare the final demonstration.

Repository reference

https://github.com/rashiyaom/SentinelCore

Assessed By (Faculty)	Marks Awarded (5 marks)	Faculty Remarks